

MECHANICAL PROPERTIES OF FLUIDS PART-3

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Pascal's Law

Pascal's law states that the pressure applied at any point of an enclosed fluid at rest is transmitted equally and undiminished to every point of the fluid and also on the walls of the container, provided the effect of gravity is neglected.

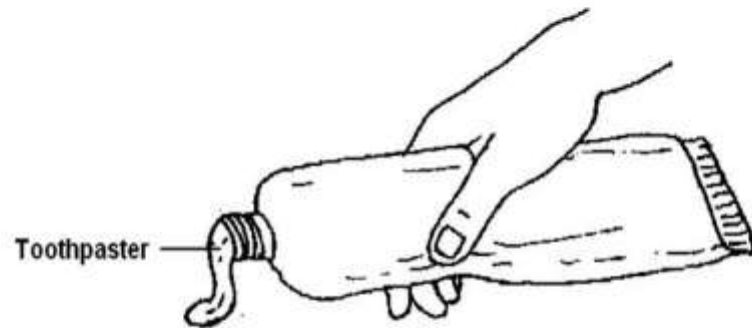
EXPLANATION OF PASCAL'S LAW

- ◉ Diagram here shows that pressure applied at top part of container is transmitted equally to every part of the container with the fluid.
- ◉ Which is shown by five pressure gauge with same readings.
- ◉ Therefore we can say that pressure acts at every point within the fluid.
- ◉ This explains Pascal's law.

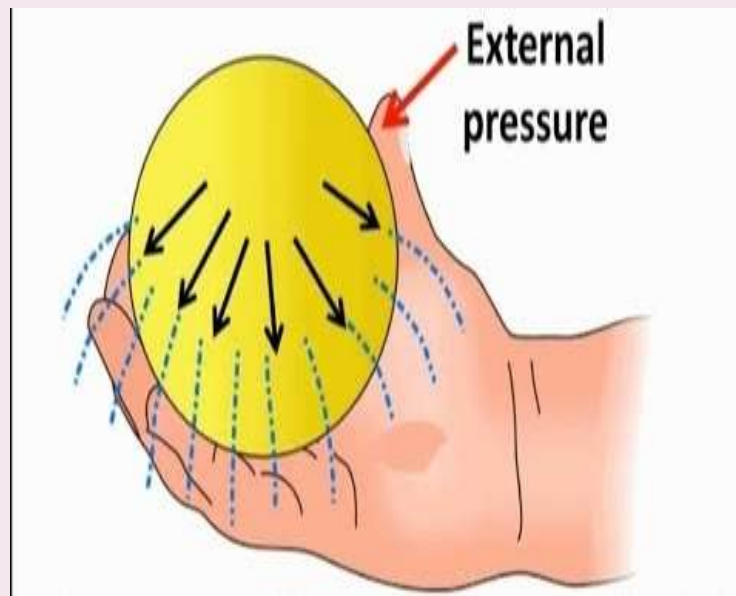


APPLICATIONS OF PASCAL'S LAW

- ◉ When you squeeze one end of the toothpaste tube, toothpaste emerges from the other end.
- ◉ The pressure has been transmitted through the fluid toothpaste.



- ◉ Pressure applied on one point of liquid transmits equally in all direction.
- ◉ This explains Pascal's law.



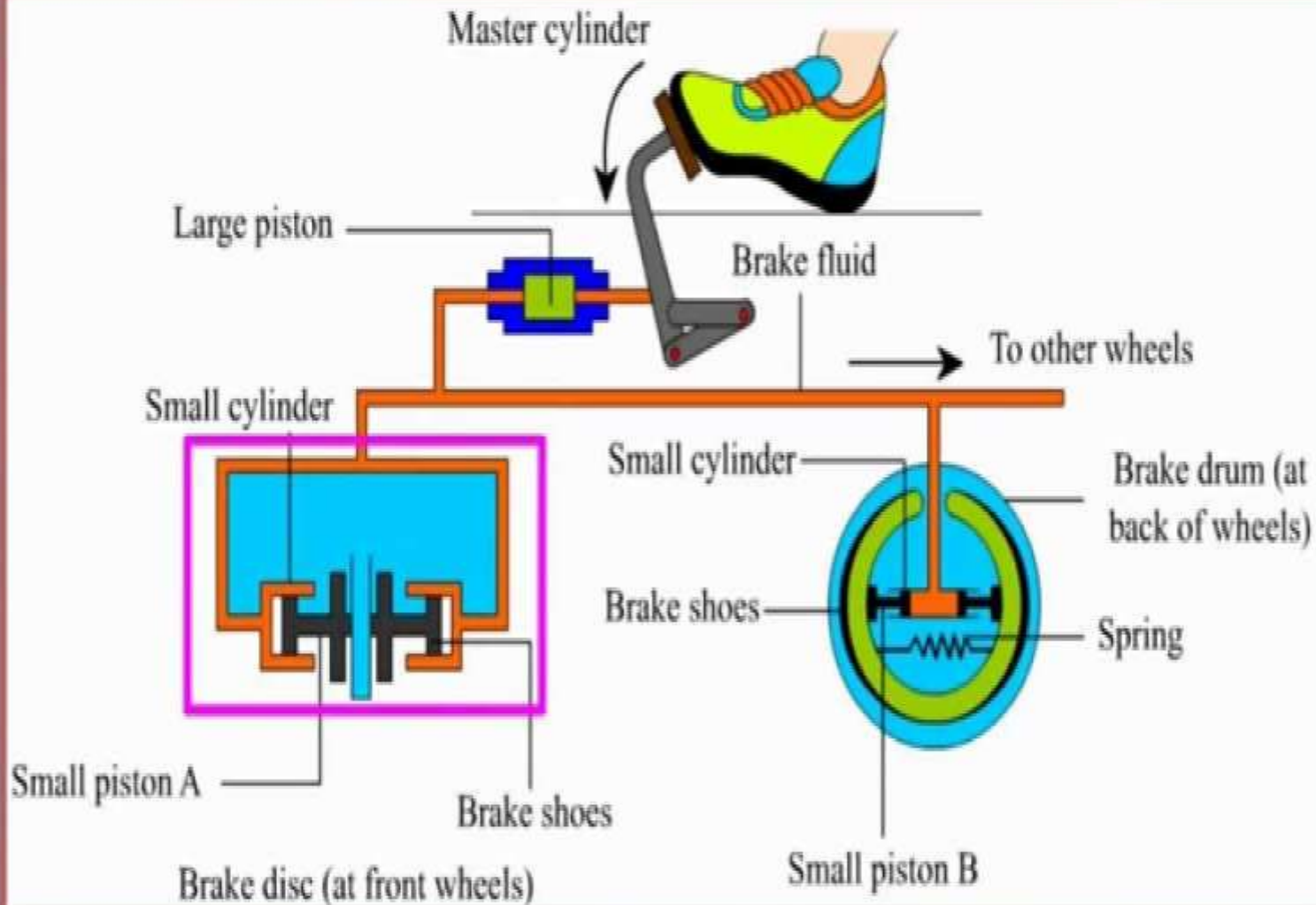
HYDRUALIC JACK



Hydraulic jack works
on the principle of
Pascal's law.

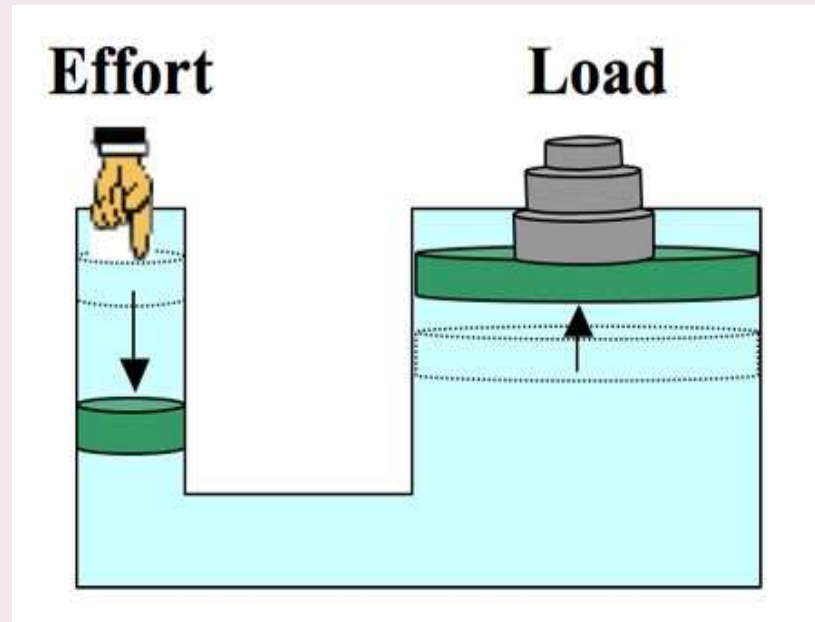
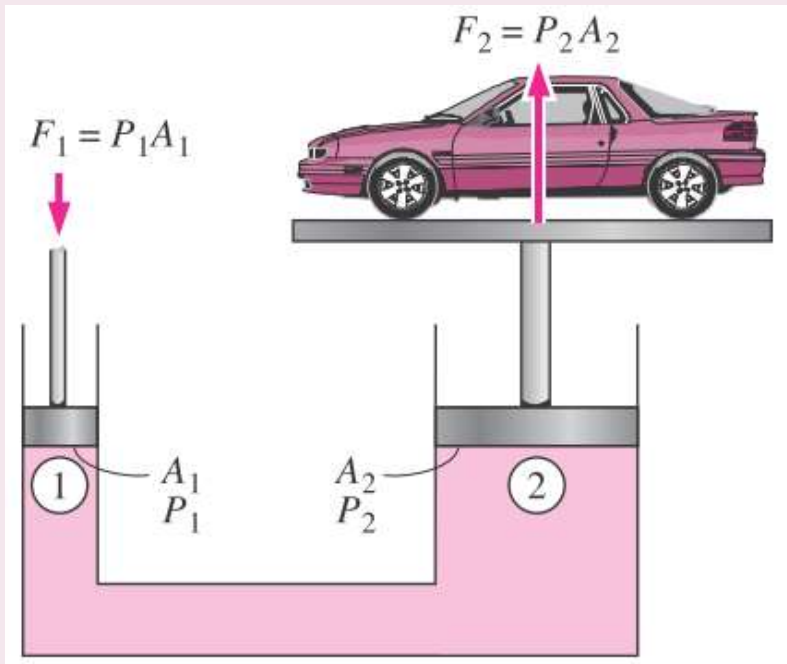
HYDRAULIC BRAKES

Hydraulic Brakes



HYDRAULIC LIFT

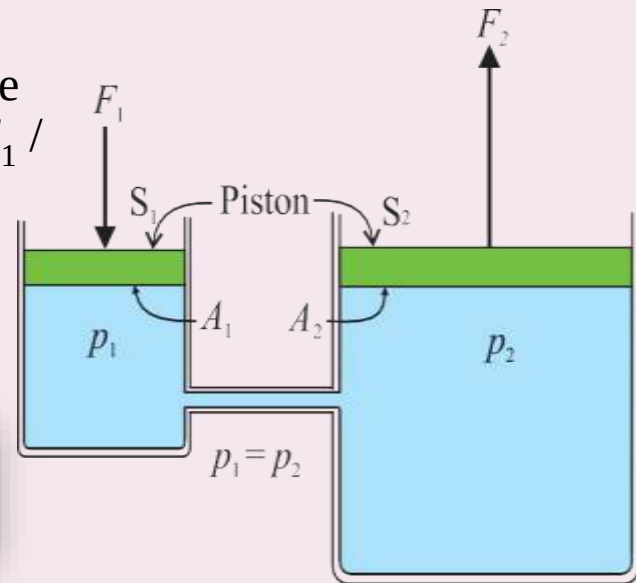
- Hydraulic lift is used to lift heavy loads like cars, trucks and bus.



WORKING OF HYDRAULIC LIFT

- Hydraulic lift is used to lift a heavy object using a small force.
- As shown in Figure, a tank containing a fluid is fitted with two pistons S_1 and S_2 .
- S_1 has a smaller area of cross section A_1 while S_2 has a much larger area of cross section A_2 ($A_2 \gg A_1$).
- If we apply a force F_1 on the smaller piston S_1 in the downward direction it will generate pressure $p = (F_1 / A_1)$ which will be transmitted undiminished to the bigger piston S_2 .
- A force $F_2 = pA_2$ will be exerted upwards on it.

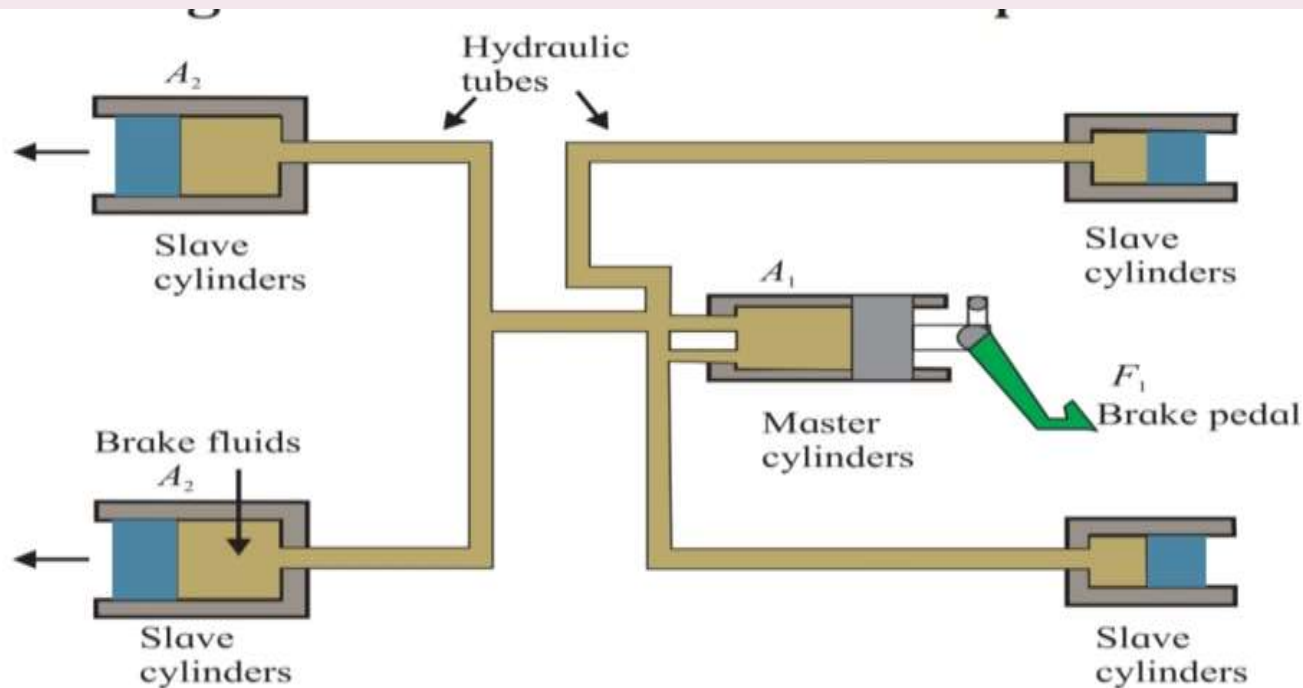
$$P_1 = P_2 \quad \rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \rightarrow \quad \frac{F_2}{F_1} = \frac{A_2}{A_1}$$



- Thus, F_2 is much larger than F_1 .
- A heavy load can be placed on S_2 and can be lifted up or moved down by applying a small force on S_1 .
- This is the principle of a hydraulic lift.

WORKING OF HYDRAULIC BRAKES

- Hydraulic brakes are used to slow down or stop vehicles in motion.
- Figure shows schematic diagram of a hydraulic brake system.
- By pressing the brake pedal, the piston of the master cylinder is pushed in forward direction. As a result, the piston in the slave cylinder which has a much larger area of cross section as compared to that of the master cylinder, also moves in forward direction.
- The slave piston pushes the friction pads against the rotating disc, which is connected to the wheel. Thus, causing a moving vehicle to slow down or stop.



- ⊙ The master cylinder has a smaller area of cross section A_1 compared to the area A_2 of the slave cylinder.
- ⊙ By applying a small force F_1 to the master cylinder, we generate pressure $p = (F_1/A_1)$. This pressure is transmitted undiminished throughout the system.
- ⊙ The force F_2 on slave cylinder is then,

$$F_2 = pA_2 = \frac{F_1}{A_1} \times A_2 = F_1 \left(\frac{A_2}{A_1} \right)$$

- ⊙ This is similar to the principle used in hydraulic lift.
- ⊙ Since area A_2 is greater than A_1 , F_2 is also greater than F_1 .
- ⊙ Thus, a small force applied on the brake pedal gets converted into large force and slows down or stops a moving vehicle.

MEASUREMENT OF PRESSURE

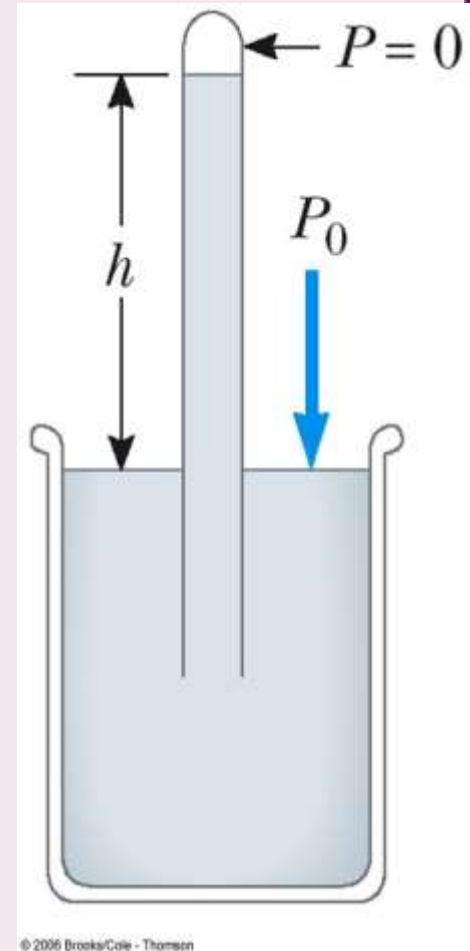
- ⦿ Instruments used to measure pressure are called pressure meters or pressure gauges or vacuum gauges.
- ⦿ Below we will describe two instruments which are commonly used to measure pressure.

MERCURY BAROMETER

OPEN TUBE MANOMETER

MERCURY BAROMETER:

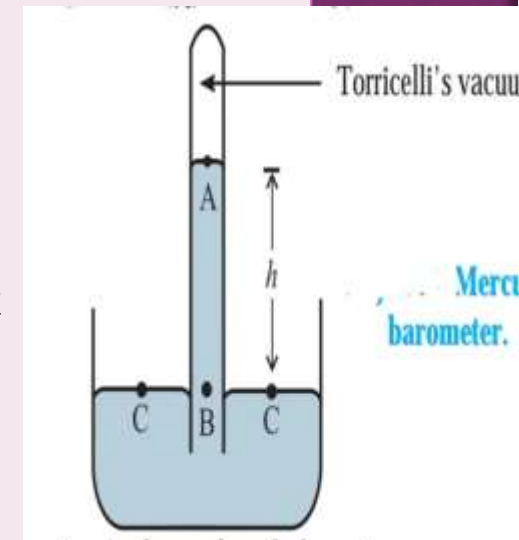
- ⦿ An instrument that measures atmospheric pressure is called a *barometer*.
- ⦿ One of the first barometers was by Italian scientist Torricelli.
- ⦿ The barometer is in the form of a glass tube completely filled with mercury and placed upside down in a small dish containing mercury.
- ⦿ Its schematic diagram is shown in Figure.



1. A glass tube of about 1 meter length and a diameter of about 1 cm is filled with mercury up to its brim. It is then quickly inverted into a small dish containing mercury.
2. The level of mercury in the glass tube lowers as some mercury spills in the dish.
3. A gap is created between the surface of mercury in the glass tube and the closed end of the glass tube. The gap does not contain any air and it is called *Torricelli's vacuum*. It does contain some mercury vapors.
4. Thus, the pressure at the upper end of the mercury column inside the tube is zero, i.e. pressure at point such as A is $p_A = \text{zero}$.
5. Let us consider a point C on the mercury surface in the dish and another point B inside the tube at the same horizontal level as that of the point C.
6. The pressure at C is equal to the atmospheric pressure p_0 because it is open to atmosphere. As points B and C are at the same horizontal level, the pressure at B is also equal to the atmospheric pressure p_0 , i.e. $p_B = p_0$.
5. Suppose the point B is at a depth h below the point A and ρ is the density of mercury then,

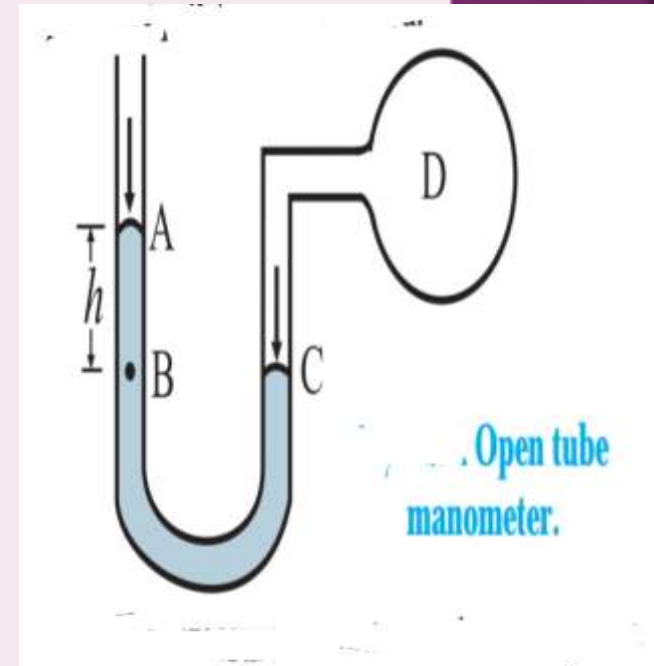
$$P_B = p_A + h\rho g$$

$p_A = 0$ (there is vacuum above point A) and $p = p$, therefore, $p = h\rho g$, where h is the length of mercury column in the mercury barometer.



OPEN TUBE MANOMETER

- ⦿ A manometer consists of a U – shaped tube partly filled with a low density liquid such as water or kerosene.
- ⦿ This helps in having a larger level difference between the level of liquid in the two arms of the manometer.
- ⦿ Figure shows an open tube manometer.
- ⦿ One arm of the manometer is open to the atmosphere and the other is connected to the container D of which the pressure p is to be measured.



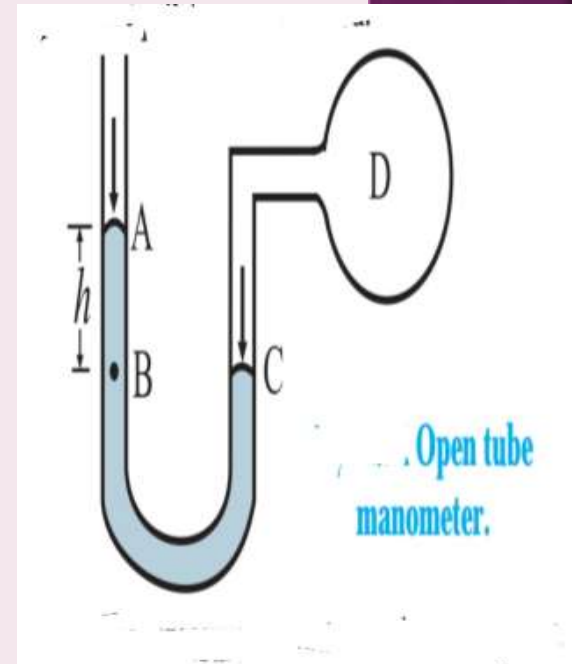
- ◉ The pressure at point A is atmospheric pressure p_0 because this arm is open to atmosphere.
- ◉ To find the pressure at point C, which is exposed to the pressure of the gas in the container, we consider a point B in the open arm of the manometer at the same level as point C.
- ◉ The pressure at the points B and C is the same, i.e.,

$$P_C = P_B \quad \dots\dots\dots(1)$$

- ◉ The pressure at the point B is

$$P_B = P_0 + h\rho g \quad \dots\dots\dots(2)$$

- ◉ where, ρ is the density of the liquid in the manometer, h is the height of the liquid column above point B, and g is the acceleration due to gravity.



⊙ According to Pascal's principle, pressure at C is the same as at D, i.e., inside the chamber. Therefore, the pressure p in the container is,

$$p = p_c$$

⊙ Using Eq. (1) and Eq. (2) we can write,

$$p = p_0 + h\rho g \quad \dots\dots(3)$$

⊙ As the manometer measures the gauge pressure of the gas in the container D, we can write the gauge pressure in the container D as

$$p - p_0 = h\rho g$$

THANK YOU